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Moving target search optimization – A literature review

Manon Raap*, Michael Preuß, Silja Meyer-Nieberg

Universität der Bundeswehr München, Fakultät für Informatik, Werner-Heisenberg-Weg 39, Neubiberg 85577, Germany



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ABSTRACT

The path-constrained moving target search problem has been subject to research since 1979 and is still investigated due to its important application, namely: search missions. This review paper provides an overview of the two main research branches. We cover approaches for path-constrained searchers, as well as approaches for kinematics-constrained searchers such as aerial search platforms. Besides an extensive overview of the literature, an overview of properties that are typical for search optimization approaches is given. A tabular summary of the approaches and their properties is provided for easy comparison and efficient lookup of applicable methods. Finally, we identify a main research gap and propose a promising type of algorithm for efficiently solving moving target search optimization problems.

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1. Introduction

Search theory has its origins in the second world war. Bernard Osgood Koopman, while working for the Anti-Submarine Warfare Operations Research Group (ASWORG), developed techniques to allocate effort to hunt German U-boats. Nowadays, applications of search theory are, e.g., the search for a life raft that is floating at sea, the search for a missing person in wilderness, or the search for radiological material in a terrorist threat scenario. Especially the rise of autonomous systems increases the importance of effective and efficient algorithms in the field of search theory. The aim of this review is to give an overview of methods that are applicable to real-life search missions. In many realistic scenarios, the object of the search mission is moving and the searcher is physically restricted to a search path. The overview in this article consists therefore of methods that account for moving targets as well as path-constrained search. A detailed scope is given in the following.

1.1. Scope

An excellent review of the search theory literature up until 1991 is described in Benkoski et al. (1991). The review at hand focuses on the problem of search for moving targets which may include resource constraints. Here, resource constraints are an abstract notion which can be specified to obstacle-, threat-, or collision avoidance and communication-, time-, or fuel constraints. We focus on discrete time approaches, whereas moving target search in continu-

ous time has been addressed in literature (Foraker et al., 2016a; 2016b; Foraker, 2011; Hibey, 1982; Ohsumi, 1984; Walton et al., 2014) as well. A recent book by Stone et al. (2016) titled 'Optimal Search for Moving Targets' bundles discrete time and continuous time approaches. Planning a search strategy can be conducted either implicitly or explicitly (Nygårds et al., 2005). In implicit planning methods, the plan specifies how the platform interacts with the environment and how it responds to sensor observations. A plan in this context is a prescription of how to react. In explicit planning methods, a search-path is computed based on predictions of the environmental state and predictions of the sensor observations. We focus on explicit planning, because it allows for anticipation of estimated target movements. Anticipation leads to a higher probability of detection compared to merely reacting to an observation.

Many related problems have been studied in literature. In order to make a clear distinction, related important problems which are not included in the scope of the research in this thesis are the following. Problems concerning terrain covering or static target search are not considered. Neither are targets considered that may react in any way to the searcher, such as evasion or cooperation. Such problems are known as two-sided search problems. We focus on one-sided search problems. Other related but excluded problems concern imaging, recognition, classification, tracking, surveillance, target state prediction, target motion modeling, sensor effectiveness, information merging, sensor fusion and shortest path to moving target problems. Nevertheless, solutions to these problems are critical to ensure the effectiveness of a search strategy in real-life search missions.

* Corresponding author.

E-mail address: manon.raap@unibw.de (M. Raap).

1.2. Contributions

The main contribution of this article is that it provides an extensive overview of approaches to solve (variations of) the moving target search problem. Furthermore, we list the properties of these approaches in a structured tabular manner, which enables an easy comparison between approaches and a quick look-up of a suitable approach as well.

1.3. Outline

The remainder of this review article is structured as follows. First, we provide an overview of the considered approaches in Section 2 and their properties in Section 3. Finally, this article concludes with a summary and a short outlook concerning promising novel research directions in Section 4.

2. Literature overview

Related literature stems from two different directions of research: operations research and applied fields such as aviation technology and robotics. The first direction focuses mainly on a high level of search strategy optimization. It aims to manage one or a team of searchers in a top-down fashion. The search area is typically divided into subareas, which are to be searched consecutively during a given number of time periods. The sequence of subareas allocated to a platform is path-constrained, i.e. consequent subareas must be adjacent. These subareas are often larger than the field of view of a sensor platform, so the actual search path *within* the subarea is not yet planned. Research from applied fields such as aviation technology and robotics focus mainly on optimizing the search path on a finer scale. Here, the resulting search path must be physically feasible for the platform. The parameters are speed, turning radius, climb/dive angle, and acceleration, which are obviously restricted by the laws of physics. In this overview, the methods from literature for path-constrained search are first described in Subsection 2.1. The methods from literature for kinematics-constrained search are described in Subsection 2.2. Finally, all discussed methods are tabularly summarized in Section 3.

2.1. Path-constrained search

The first approaches under review are those optimizing a path-constrained search strategy. Within this category, the methods are divided into exact methods for a single searcher, heuristics for a single searcher, and methods (both exact and approximate) for multiple searchers.

Exact methods for a single searcher

Eagle (1984b) was the first to consider the problem as a partially observable Markov decision process, assuming Markovian target motion. As it was standard for problems exploiting a Markov assumption, the solution technique here is dynamic programming. Possible search paths are defined by a finite set of vectors. Eagle noted that some vectors are dominated and can be removed from consideration. He proposed a potentially large linear program to check for dominance, supplemented by two alternative heuristic reduction schemes. Stewart (1979) introduced the path-constrained search problem in 1979. He chose a depth-first branch & bound approach, because most approaches to linear binary problems were of this type. In this algorithm, lower bounds on the probability of non-detection are obtained by relaxing the searcher's path constraints. The resulting substitute problems are being solved using a modification of Brown's algorithm Brown (1980). However, Brown's algorithm does not guarantee optimality for discrete search effort, so obtained bounds are only approximate. This

may result in mistakenly pruning an optimal branch. An experimental analysis with this method is presented by Stewart (1980). Eagle and Yee (1990) continued the work by replacing the approximate bound with a true lower bound, which is obtained by relaxing the integrality constraints of the searcher's position. The relaxed problem is then solved using the Frank-Wolfe method Frank and Wolfe (1956).

Martins (1993) reconsidered the Eagle-Yee procedure and proposed a new bound. Contrary to Stewart's and Eagle-Yee's approaches, no constraints are relaxed. Instead, the expected number of detections provides an upper bound for the maximum probability of detection. This objective function is linear in the decision variables, so maximizing the expected probability of detection is equivalent to finding the longest path through a directed acyclic network. The longest path algorithm is adapted from Cormen et al. (2009), which has linear complexity. This bound is more easily evaluated than the Eagle-Yee bound, but is less tight. Despite more branching, run times are still roughly a factor of four smaller on all test problems.

In 1995, Washburn summarized the computational experiences of the branch & bound procedures mentioned above in Washburn (1995). He proposed and evaluated a type of hybrid approach where a backup bound is calculated if and only if the primary bound comes within δ of fathoming the segment, where $\delta \geq 0$. This arises from the idea that putting in some more effort of computing a tighter bound may prevent branching. In all examined test cases, a pure procedure is at least as good as any hybrid. A hybrid method is generally slightly slower, but Washburn suggests that the gain in robustness might be worth the sacrifice. Washburn's preferred hybrid procedure is Martins' bound backed up by the Eagle-Yee bound.

Washburn (1998) published another paper on computational results on several bounds in the branch & bound procedure, a few years later in 1998. He proposed a new bound, which combines the relaxation of the integrality constraint on the searcher's position and a linear overestimation of the glimpse probability function. The need for solving a longest path problem is removed, but the bound is less tight on account of the additional relaxation. His iterative forward-and-backward algorithm Washburn (1983) yields optimal results for this relaxed problem.

Lau et al. (2008) introduced an improvement of Martins' bound, called the discounted mean (DMEAN) bound. In the calculation of Martins' bound, the reward at time k is calculated while ignoring all observations made at time steps other than k . It is known that the true probability of detection at time step k depends on the cumulative probability of non-detection at all previous time steps. By additionally accounting for the probability of detection at time step $k - 1$, the bound is tightened. They showed with the help of computational analyses that this bound leads to faster solution times than any of the existing bounds, because it is significantly tighter with almost no additional computation costs. The approach was furthermore extended to find solutions for the path-constrained search problem with non-uniform travel times.

All bounds mentioned so far are *dynamic* in the sense that they are optimized before each branching. Sato (2008), on the other hand, introduced *static* (SB) versions of Martins' MEAN bound and Lau's DMEAN bound. These bounds are weaker, but have the advantage that the longest path is calculated only once prior to any branching, instead of before each branching attempt. Especially the latter *directional* static bound leads to shorter run times. He furthermore proposed a network reduction method. Relocation arcs are generated, leading towards nodes-of-first-contact where the searcher has the earliest possibility of detection. This procedure may reduce the amount of branching attempts significantly.

Morin et al. (2009) extended the path-constrained search problem by adding an inter-region visibility criterion. In this new for-

mulation, a searcher is able to make one observation in any of the cells adjacent to its position. They proposed a novel mixed-integer linear programming (MILP) problem, the instances of which are solved using a commercial solver.

Hohzaki and Iida (1997) considered an extension to the path-constrained search problem, where the decision of whether or not to look has to be made for each time period additionally. A look does not only yield a reward by increasing the probability of detection, but it also comes with a cost. In their approach to obtain an upper bound, the path-constraints as well as the integrality constraints on the searcher's position are relaxed. Additionally, they relax the integrality constraints on the look decisions. The resulting problem is then solved using Washburn's forward-and-backward algorithm Washburn (1983) and incorporated in a branch & bound procedure.

Sato and Royset (2010) extended Sato's original approach (Sato, 2008) by constraining the resources of the platform. They considered multiple constraints on the consumption of resources such as time, fuel, and risk. In this problem, the glimpse probability is assumed to be history dependent. They propose a branch & bound method, with a Lagrangean directional static bound on the optimal probability of detection.

Heuristics for a single searcher

Besides exact methods, many researchers consider at least one heuristic method due to the intractability of the overall problem for large instances. Using a receding horizon approach was first proposed by Eagle (1984a) where he applied it to his dynamic programming method. Martins (1993) introduced a heuristic method besides his exact approach. For each time period k , the search-path is extended by maximizing the expected probability of detection (ED) in times $k, \dots, K$. The probability map, given that no detection has occurred, is updated after each extension. He uses this heuristic to effectively derive a good initial solution for his branch & bound procedure. Washburn (1998) also proposed a heuristic to find a nearly optimal solution, rather than exactly optimal. Instead of branching when the lower bound q is smaller than the best (lowest) non-detection probability q^* found so far, branching is carried out when $q + d < q^*$. The last saved path will be within d of optimality. Increasing d from 0.01 to 0.02 already resulted in an order of magnitude decrease in runtime. Washburn saw this heuristic as perhaps being the best hope for solving search problems of a practical size. Thomas and Eagle (1995) considered approximating bounds by procedures that are more easy to calculate. For their stationary target approximation all target states are assumed to be recurrent and for their no-learning approximation the transition matrix must have identical rows.

Sato (2008) proposed a static bound heuristic (SBH) similar to Martins' ED heuristic, but adjusted with his own static bound. As a result, it performs only one longest path calculation. However, SBH is not available for huge problems because the time-expanded network becomes too large for computations.

Hong et al. (2009a) introduced the depth- l approximation, which is similar to Sato's SBH. A time-expanded network is constructed and named depth- l network, because the cost of an arc is computed from the probability of non-detection over the previous l time steps. The second step of the algorithm is to solve the shortest path on this acyclic network, which requires linear time. Sato's SBH would be a depth-2 approximation in this setting. The computation costs increase exponentially with l .

Lanillos et al. (2012) applied cross entropy optimization, where samples are generated using multinomial sampling. They evaluated the path-constrained search by several metrics: the mean time to detection and the time-discounted probability of detection. The resulting probability of detection of the search paths when optimiz-

ing those metrics is compared, where the latter dominates on almost all test instances.

Morin et al. (2012) presented a constraint programming model. The proposed value selection heuristic simplifies the probability system by ignoring past non-detection events. At each time step, the accessible vertex is chosen that maximizes the total probability of detection in the remaining time, not conditioned on non-detection in previous time steps. The authors pointed out that the main advantage of this approach is expressiveness, because the model stays close to the natural problem formulation. Morin et al. (2010) furthermore proposed a metaheuristic, i.e., an ant colony optimization, method to solve their MILP with an inter-region visibility criterion. Simulations showed promising results for relatively large instances.

Stewart (1985) considered the realistic case that a target may leave behind some evidence of its former presence. He proposed to optimize search by actively searching for a detectable trail left behind by a target. A moving-horizon rule and a heuristic simplification thereof were introduced.

Hong et al. (2009b) expanded their depth- l network (Hong et al., 2009a) so that it accounts for a multiple-search-speed option, whereby a trade-off between speed and glimpse probability is made. The set of reachable cells within one time period is extended. This causes the number of arcs in each layer of the time-expanded network to increase significantly with the searcher's speed. They again used a longest path computation on the depth- l network.

Approaches for multiple searchers

Dell et al. (1996) concentrated on solving the path-constrained search problem with multiple searchers. They revisited Martins' procedure, which was found to exceed the acceptable computation time when applied to the multi-searcher problem. Several heuristics are presented, of which two exhibit reasonable run times and quality solutions. These heuristics are both centralized methods for cooperation. The first one extends Martins' ED heuristic for multiple searchers. It obtains solutions within 2% of the best known solution for each of the one- and two-searcher test problems considered. The second metaheuristic is a genetic algorithm, which was efficient on the three searcher problems, but no guarantee on the solution quality can be given.

To optimize the path-constrained search problem with multiple searchers, Sato (2008) introduced a time-expanded configuration network. A configuration represents the combined locations of all searchers. He then applied his SB method to this network, which is tractable only for very small instances due to the large number of arcs. Therefore two heuristics were introduced as alternatives. The first is to apply the SBH on the time-expanded configuration network. Sato pointed out that this heuristic is not applicable to large problems because the time-expanded configuration network cannot be generated due to computational limits. The second heuristic is a cross entropy optimization method, which performs well on instances with as many as 30 searchers.

A method called the search and rescue optimal planning system is actually being used by the United States Coast Guard and is described by Kratzke et al. (2010). The technology behind it stands out compared to all other approaches with respect to the planning method. Here, a search-path is a pattern of equally spaced parallel paths within a rectangle. Instead of allocating a platform to a cell, the size, shape, and position of a rectangular subarea are optimized. The algorithm plans heuristically by placing a standard shaped rectangle on a cell with the highest probability, performs an accordion search (Kratzke et al., 2010) to find the proper size and location, and then fine-tunes it while minimizing overlap. A generalization of the multi-searcher problem was presented by Royset and Sato (2010). They introduced a novel MINLP for the

problem with multiple targets, searcher deconfliction and target and location dependent search effectiveness. The objective is to minimize the largest non-detection probability. The authors suggested using a cutting plane method, which results in a practical exact algorithm and is applicable for both conditionally deterministic and Markovian target models. This approach performs strongly compared to a branch & bound method, which fails on all test instances examined. A main advantage is the scalability as the number of searchers grows.

2.2. Kinematics-constrained search

The following approaches all aim for optimizing kinematics-constrained search taking the kinematical properties of the searcher into account. The overview starts by describing the publications considering kinematics-constrained search for a single searcher, followed by approaches for a team of searchers.

Approaches for a single searcher

The first approach explicitly considering flight kinematics was presented by Bourgault et al. (2006). The authors used a discrete-time non-linear velocity model, which takes a limited turn rate and a limited speed as input. Simulations show the effectiveness of a myopic and of a piecewise-constant control implementation. Furukawa et al. (2007) propose a new method for a continuous representation of the probability map: the element-based method. For planning, they utilized a similar myopic approach to Bourgault et al. (2006), however uniquely using the Kullback–Leibler divergence as the objective function. Collins et al. (2007) were the first and, to the authors' knowledge, only ones to propose a planning method that plans for both a search path and sensor schedule within the scope of moving target search. A unique dual probability map representation is used, combining the particle filter and grid-based representations. They furthermore considered a dynamic graph whose nodes are placed at high-reward locations and are dynamically updated at each time step. For a combined kinematics-constrained search path and sensor schedule planning, they suggested two implementations. The first of these is by evaluating search paths based on the cumulative probability of containment within the field of regard. The second is by evaluating search paths based on optimization of the sensor schedule on the path. Raap et al. (2017b) propose a mixed integer linear programming formulation to optimize kinematics-constrained search over a fixed number of time steps.

Approaches for a team of searchers

The following approaches consider a cooperative search team, where historical and planned observations are exchanged and adapted to achieve a non-redundant search strategy.

Bourgault et al. (2003) extended their work from Bourgault et al. (2006) and suggested a coordinated approach for multiple heterogeneous searchers, which are considered to behave in a completely decentralized manner. They found that a myopic form of coordinated search strategy provides very sensible control solutions at low computational costs. Hu et al. (2012) proposed a gradient based control law, where the optimal next-time sampling position is chosen to maximize the information gathered at time k . This is a myopic heuristic with a simple best response type of cooperation. Another contribution is their distributed probability map updating model, assuming that each platform only communicates with its neighbors within communication range. In this model, the glimpse probability varies as a function of sensor altitude. Xiao et al. (2012) applied a hybrid method. Its main idea is that a virtual force type method helps the receding horizon method to improve search efficiency and reduce the computational costs. This method avoids the problem of distinguishing

between paths in low uncertainty areas arising in receding horizon methods.

An optimal team reward can be achieved either in a centralized manner, or in a decentralized manner using negotiation. Wong et al. (2005) propose a centralized method for cooperation, where the cumulative team reward is maximized. To overcome the problem of distinguishing between paths in low uncertainty areas, they introduced a switching objective function to direct the searcher towards uncertainty areas. In this case, the objective is to minimize the distance to the nearest uncertainty area. Aiming for a team-optimal reward, while avoiding centralized computation, Bourgault et al. (2004) propose a decentralized negotiation scheme. They use a block-iterative non-linear algorithm. This consists of iteratively fixing all paths, except for the i th path which is then optimized with respect to the fixed paths. At termination, the search plan satisfies Nash's equilibrium. A first method aiming for optimal cooperation in a decentralized setting was presented by Delle Fave et al. (2010). They suggested using a max-sum algorithm to optimize the joint probability of detection, which operates over a factor graph. This approach scales linearly with the numbers of searchers. Cole (2009) presented an entire cooperative unmanned aerial system architecture for information theoretic searching and tracking. In a search for an unknown number of targets, which is Poisson distributed, the expected number of detected targets is maximized.

The objective of minimizing Shannon's entropy is applied in simulations in Peng et al. (2015). A method based on enumeration of all possible strategies is used and compared against, amongst other strategies, the Zamboni strategy of fully covering the search area in rectangular search strips. A combined strategy for autonomous take off, search, and landing for a team of UAVs is presented in Meng et al. (2016). Here, a simple myopic algorithm is used for waypoint planning. When a UAV arrives at a crossroad, the next waypoint is set to the end of a road which is currently not searched by other UAVs and has the highest probability of containment.

Perez-Carabaza et al. considered minimal time search problems with several unmanned aerial vehicles (Perez-Carabaza et al., 2017; 2016). In Perez-Carabaza et al. (2016), they introduced a detailed UAV motion model that was implemented in MATLAB and used in Perez-Carabaza et al. (2017) where mobile targets were considered. The planner employed a variant of a continuous ant colony optimization algorithm, the so-called ACO_R. It uses an archive of best solutions and samples new solutions based on normal distributions. The distribution parameters are obtained with the help of the archive. In addition a heuristic was introduced which biases the solution towards the areas with high probability values. The authors compared two ACO-variants as well as an evolutionary algorithm on five scenarios. Based on the approximation of the expected time until target detection, they identified advantages for the ACO_R variant that uses the heuristic.

3. General principles of search methods

All discussed approaches are summarized in Tables 1 and 2. In these tables, the most important properties of the approaches with respect to assumptions and solutions are presented in the columns. Each property is abbreviated to fit in a table cell. All properties are elucidated first, starting with the motion model of the target. When a certain property has not been specified in the associated article, the property is denoted by the entry (-).

The first property concerns the motion model of the **target**. In each model, targets are assumed to be independent. The generic (G) target model consists of a set of possible target tracks, where each possible track has a given probability of the target taking that track. The number of possible tracks grows exponentially in time

Table 1
Overview of approaches for path-constrained search.

Reference	Year	Target motion	Sensor model	Composition	Cooperation	Objective	Additional constraints	Additional output	Solution quality	Formulation	Algorithm type	Horizon	Speed Trade-off	Searcher State Space
Path-constrained search														
Exact methods for single searcher														
Eagle Eagle (1984b)	1984	M	I/1/St	Si	–	PND	–	–	O	POMDP	DP	K	–	FD
Stewart Stewart (1979)	1979	G	I/1/St	Si	–	PD	–	–	A	MINLP	BB	K	–	FD
Eagle and Yee Eagle and Yee (1990)	1990	M	I/1/St	Si	–	PD	–	–	O	MINLP	BB	K	–	FD
Martins Martins (1993)	1993	M	I/1/St	Si	–	PD	–	–	O	MINLP	BB	K	–	FD
Washburn Washburn (1998)	1998	M	I/1/St	Si	–	PND	–	–	O	MINLP	BB	K	–	FD
Lau <i>et al.</i> Lau et al. (2008)	2008	M	I/1/St	Si	–	PD	–	–	O	MINLP	BB	K	–	FD
Sato Sato (2008)	2008	M	I/1/St	Si	–	PD	–	–	A	MINLP	BB	K	–	FD
Morin <i>et al.</i> Morin et al. (2009)	2009	M	I/1/Gi	Si	–	PD	–	Or	O	MILP	CS	K	–	FD
Hohzaki and Lida Hohzaki and Lida (1997)	1997	G	I/1/St	Si	–	PD	LR	Lo	O	MINLP	BB	K	–	FD
Sato and Royset Sato and Royset (2010)	2010	M	I/1/St	Si	–	PD	LR	Alt	O	MINLP	BB	K	RA	FD
Heuristics for a single searcher														
Eagle Eagle (1984a)	1984	M	I/1/St	Si	–	PND	–	–	A	POMDP	DP	K	–	FD
Martins Martins (1993)	1993	M	I/1/St	Si	–	PD	–	–	A	MINLP	LP	Myopic	–	FD
Washburn Washburn (1998)	1998	M	I/1/St	Si	–	PND	–	–	A	MINLP	BB	K	–	FD
Thomas and Eagle Thomas and Eagle (1995)	1995	M	I/1/St	Si	–	PD	–	–	A	MINLP	BB	K	–	FD
Sato Sato (2008)	2008	M	I/1/St	Si	–	PD	–	–	A	MINLP	LP	Myopic	–	FD
Hong <i>et al.</i> Hong et al. (2009a)	2009	M	I/1/St	Si	–	PND	–	–	A	–	LP	K	–	FD
Lanillos <i>et al.</i> Lanillos et al. (2012)	2012	M	I/1/St	Si	–	DTR	–	–	A	OC	CE	K	–	FD
Morin <i>et al.</i> Morin et al. (2012)	2012	M	I/1/Gi	Si	–	PD	–	Or	A	CP	CS	K	–	FD
Morin <i>et al.</i> Morin et al. (2010)	2010	M	I/1/Gi	Si	–	PD	–	Or	A	–	ACO	K	–	FD
Stewart Stewart (1985)	1985	G	I/1/St	Si	–	PD	–	–	A	POMDP	BB	K	–	FD
Hong <i>et al.</i> Hong et al. (2009b)	2009	M	I/1/St	Si	–	PD	–	Sp	A	–	LP	K	RE	FD
Approaches for multiple searchers														
Dell <i>et al.</i> Dell et al. (1996)	1996	G	I/1/St	He	Ce	PD	–	–	A	MINLP	Enum	Myopic	–	FD
Dell <i>et al.</i> Dell et al. (1996)	1996	G	I/1/St	He	Ce	PD	–	–	O	MINLP	BB	K	–	FD
Sato Sato (2008)	2008	M	I/1/St	He	Ce	PD	–	Alt	A	MINLP	BB	K	–	FD
Sato Sato (2008)	2008	M	I/1/St	He	Ce	PD	–	Alt	A	MINLP	CE	K	–	FD
Sato Sato (2008)	2008	M	I/1/St	He	Ce	PD	–	Alt	O	MINLP	LP	K	–	FD
Kratzke <i>et al.</i> Kratzke et al. (2010)	2010	G	I/N/St	He	Ce	PD	–	–	A	–	–	K	–	FD
Royset and Sato Royset and Sato (2010)	2010	G	I/1/St	He	Ce	PND	CA	–	O	MINLP	OA	K	–	FD
Royset and Sato Royset and Sato (2010)	2010	G	I/1/St	Ho	Ce	PND	CA	–	O	MILP	CS	K	–	FD
Royset and Sato Royset and Sato (2010)	2010	M	I/1/St	Ho	Ce	PND	CA	–	O	MILP	CS	K	–	FD

Table 2
Overview of approaches for kinematics-constrained search.

Reference	Year	Target motion	Sensor model	Composition	Cooperation	Objective	Additional constraints	Additional output	Solution quality	Formulation	Algorithm type	Horizon	Speed Trade-off	Searcher State Space
Kinematics-constrained search														
Approaches for a single searcher														
Bourgault <i>et al.</i> Bourgault et al. (2006)	2006	M	I/N/St	Si	–	MTD	–	–	A	OC	Enum	Myopic	–	IC
Furukawa <i>et al.</i> Furukawa et al. (2007)	2007	M	I/N/St	Si	–	KLD	–	–	A	OC	–	Myopic	–	IC
Collins <i>et al.</i> Collins et al. (2007)	2007	M	I/N/Gi	Si	–	PD	–	Or	A	OC	–	K	RA	FD
Raap <i>et al.</i> (2017a, 2017b)	2017	M	I/N/St	Si	–	PD	–	–	O	MILP	CS	K	–	FD
Approaches for a team of searchers														
Bourgault <i>et al.</i> Bourgault et al. (2003)	2003	M	I/N/St	He	Co	MTD	–	–	A	OC	–	Myopic	–	IC
Hu <i>et al.</i> Hu et al. (2012)	2012	R	I/N/St	He	Co	PD	–	Alt	O	OC	GM	Myopic	–	IC
Xiao <i>et al.</i> Xiao et al. (2012)	2012	R	I/N/St	He	Co	PD	CA	–	A	OC	VF	Myopic	OF	IC
Wong <i>et al.</i> Wong et al. (2005)	2005	M	I/N/St	He	Ce	PPD	–	–	A	OC	–	Myopic	OF	IC
Bourgault <i>et al.</i> Bourgault et al. (2004)	2004	M	I/N/St	He	Ne	PD	–	–	A	OC	–	Myopic	–	IC
Fave <i>et al.</i> Delle Fave et al. (2010)	2010	M	I/N/St	He	Ne	PD	–	–	A	OC	–	Myopic	–	ID
Cole Cole (2009)	2009	M	I/N/St	He	Co	EN	–	–	A	OC	–	Myopic	–	IC
Peng <i>et al.</i> Peng et al. (2015)	2015	R	–	He	Co	SE	–	–	O	OC	Enum	K	–	FD
Meng <i>et al.</i> Meng et al. (2016)	2016	DC	I/1/St	He	Co	PD	–	–	O	–	–	Myopic	–	FD
Perez-Carabaza <i>et al.</i> Perez-Carabaza et al. (2017)	2017	DC	I/N/-	Ho	Ce	MTD	CA	–	A	OC	ACO, GA	K	–	IC

and space. Monte Carlo sampling methods can be used to generate a finite set of possible tracks and their corresponding probabilities. The Markovian (M) target model is more restricted. It assumes that the process has the Markov property (Norris, 1999), so that the position of the target at time $k + 1$ only depends on its current position at time k and is independent of those at any previous point in time. The random (R) target model is a special case of the Markovian target model. Here, the target moves to one of the cells that are adjacent to its current cell with equal probability. Such a target is also said to be *diffusing*. In case no assumption on the target motion can be made, the model of decaying certainty (DC) can be used, in which the probability of containment of a cell increases over time after it has been observed. A deterministic (D) model can be used when the target track is assumed to be known.

Each searcher employed is equipped with a **sensor**. Sensors for detection are e.g. electro-optics/infra-red sensors. Several aspects of such sensors are important. These are sensor range, agility, and performance, which are therefore considered in detail within literature. These properties are presented in a model of the form range/performance/agility. The range of a sensor is either assumed to be restricted to the cell (1) which is currently observed by its carrying platform, or restricted to a predefined range (N). We distinguish between perfect (P) and imperfect (I) sensors with respect to their performance. A perfect sensor detects a target within its field of view with probability one, whereas an imperfect sensor may miss the target. For sensor agility, we distinguish between stabilized (St) and gimballed (Gi) sensors.

The team **composition** can be made up of a single (Si) searching platform or a team of multiple platforms searching cooperatively. These platforms may differ in general characteristics concerning speed, turn radii, and sensor quality, in which case the team is heterogeneous (He) and it is homogeneous (Ho) when the team is uniform.

Cooperation within a team can be achieved in several ways. Using a centralized (Ce) control, an optimal search strategy is found by solving one overall optimization problem for the entire team. Computation is unfortunately often only tractable for very small instances. A decentralized approach is often less computationally expensive and is also less prone to loss of communication. Cooperation means that observations and planned search paths are exchanged and adapted to aim for optimal team search results. Two types of cooperation exist, where the difference lies in the balance between optimality of team results and computational costs. The cheapest type of cooperation from a computational point of view is coordination (Co). The predicted observations on the optimized search paths are exchanged between the platforms. The historical and predicted measurements are fused into the (shared or individual) probability map and the platform specific problems are optimized accordingly in sequence. Additional negotiation (Ne) between platforms improves the objective value, but increases computational costs and communication effort in general.

The quality of a search strategy must be defined in terms of an **objective**. Several types of objectives have been subject to studies. Which type to use depends on the goal of the search mission, and therefore they are incomparable in terms of effectiveness. The objective value is typically called the *search reward* within the context of target search. Maximizing the probability of detection (PD) is effective when a fixed time frame for the search mission is prescribed. The decision maker wishes to maximize the probability of detection cumulative over all time periods. Minimizing the probability of non-detection (PND) (or *overlook* probability) is the opposite of maximizing the probability of detection. Maximizing the time-discounted reward (TDR) discounts the reward of probability of detection in each time step. Alternatively, the mean time to detection (MTD) can be considered, or the expected number of detec-

tions (EN) in the case of a large number of targets. The information entropy (or Shannon's entropy) (SE) (MacKay, 2002) is a measure of uncertainty in which higher values indicate higher uncertainty of the whereabouts of the target. The Kullback–Leibler divergence (KLD) Kullback and Leibler (1951) is another measure based on information, described as the relative entropy between the start and the end of the search mission.

Additional constraints with respect to the search strategy can exist. The search path or strategy can be subject to limited resources (LR), such that the search objective must be optimized while not exceeding the allowed resource consumption. Examples of resources accounted for in literature are: fuel, observations, and risk exposure. Some approaches explicitly account for collision avoidance (CA).

Additional outputs to the waypoints or allocations can be e.g. altitude (Alt) of the platform at a time step, whether or not to make an observation (Lo), the speed of the platform (Sp), its assignment to alternative tasks (AT) such as refueling or rescue, or the orientation of the sensor (Or).

A **speed trade-off** can be modeled in case the speed of the platform can be increased for quicker relocation at the cost of decreased glimpse probability. A few approaches propose methods to balance search reward and relocation speed. Three types of search-relocation trade-off methods can be distinguished. A switch between objective functions or a switching objective function (OF) can be used, alternating between minimizing the distance to the nearest uncertainty area and maximizing the search reward. The decision between these objectives is based on an adaptive switching parameter, which may be platform specific. Another type of method is to extend the range (RE) of reachable cells or waypoints within a single time period to directly adjacent cells (direct neighbors). By increasing the speed, the neighborhood is expanded. This enlarges the number of possible actions at each time period, and thereby the number of possible search paths significantly. Relocation arcs (RA) can be used when the platform state space is modeled as a (time-expanded) network. It is then possible to determine a set of nodes-of-first-contact. These are the nodes at which the searcher can detect the target the earliest. The arcs leading from the searcher's current position towards these nodes are called relocation arcs and take multiple time periods to traverse. To enable solution approaches, the platform state space is either an infinite continuous (IC) state space or a finite discrete (FD) state space in the form of a grid. Based upon the platform state space, the search problem is formulated in literature as either a partially observable Markov decision process (POMDP), a mixed integer linear program (MILP), a mixed integer non-linear program (MINLP), a constraint program (CP), or an optimal control (OC).

There are different **types of algorithms** to solve a search strategy planning problem formulation. The following exact algorithm types and metaheuristics have been applied in the reviewed literature. The exact algorithm types are dynamic programming (DP), branch & bound (BB), outer approximation (OA), and constraint programming (CP). Metaheuristics applied to the path-constrained search problem are genetic algorithms (GA), ant colony optimization (ACO), and cross entropy (CE). A virtual force field algorithm (VF) is a typical robot guiding algorithm. Other algorithm types are the gradient method (GM) and the longest path on a directed acyclic graph (LP) (Cole, 2009). A few approaches do not propose a new algorithm, but deploy a commercial solver (CS) to solve the problem formulation. The least efficient algorithm type is to enumerate (Enum) over all feasible search strategies.

Finally, optimal (O) or approximate (A) **solution qualities** are proposed for a certain **planning horizon**. Planning for the full duration (K) of a predefined number of time periods K leads to (approximation of) globally optimal search strategies, when computation is tractable. A myopic (Myopic) approach chooses the platform

state that is optimal for the next time period, possibly with a cost-to-go approximation (Bertsekas, 2012).

4. Conclusion and outlook

In this literature review on moving target search optimization in discrete time and space, we discussed 30 methods for path-constrained search and 14 methods for kinematics-constrained search for, e.g., aerial search platforms. The provided tables give a complete but concise overview of the existing methods and their properties. This allows for a quick comparison between approaches, as well as easy lookup of a method that is applicable to specific real-life search missions. One of the main research gaps that stand out after analysis of this literature overview, is the moving target search problem for a team of searchers with limited resources and accounting for the kinematical constraints of the platforms. Due to the underlying structure of this problem, namely the combinatorial nature of the trajectories of the searchers, a Dantzig-Wolfe Decomposition approach appears to be very promising for solving this problem efficiently. This approach is the subject of our further research.

References

- Benkoski, S.J., Monticino, M.G., Weisinger, J.R., 1991. A survey of the search theory literature. *Nav. Res. Logist.* 38 (4), 469–494.
- Bertsekas, D.P., 2012. *Dynamic Programming and Optimal Control*, 2, 4th Edition. Athena Scientific.
- Bourgault, F., Furukawa, T., Durrant-Whyte, H., 2003. Coordinated decentralized search for a lost target in a Bayesian world. In: *Proceedings of the 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Vol. 1, pp. 48–53.
- Bourgault, F., Furukawa, T., Durrant-Whyte, H., 2006. Optimal search for a lost target in a Bayesian world. In: Yuta, S., Asama, H., Prassler, E., Tsubouchi, T., Thrun, S. (Eds.), *Field and Service Robotics*, Vol. 24 of Springer Tracts in Advanced Robotics. Springer, Berlin Heidelberg, pp. 209–222.
- Bourgault, F., Furukawa, T., Durrant-Whyte, H., 2004. Decentralized Bayesian negotiation for cooperative search. In: *Proceedings of the 2004 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Vol. 3, pp. 2681–2686.
- Brown, S.S., 1980. Optimal search for a moving target in discrete time and space with an exponential detection function. In: Haley, K.B., Stone, L.D. (Eds.), *Search Theory and Applications*, Vol. 8 of NATO Conference Series. Springer, US, pp. 221–229.
- Cole, D.T., 2009. *A Cooperative UAS Architecture for Information-Theoretic Search and Track*. University of Sydney Ph.D. thesis.
- Collins, G.E., Riehl, J.R., Vegdahl, P.S., 2007. A UAV routing and sensor control optimization algorithm for target search. In: *Proceedings of the 2007 Defense and Security Symposium*, International Society for Optics and Photonics. pp. 65610D–65610D.
- Cormen, T.H., Leiserson, C.E., Rivest, R.L., Stein, C., 2009. *Introduction to Algorithms*. MIT Press, Cambridge, MA, USA.
- Dell, R.F., Eagle, J.N., Martins, G.H.A., Santos, A.G., 1996. Using multiple searchers in constrained-path, moving-target search problems. *Nav. Res. Logist.* 43 (4), 463–480.
- Delle Fave, F.M., Xu, Z., Rogers, A., Jennings, N.R., 2010. Decentralised coordination of unmanned aerial vehicles for target search using the Max-Sum algorithm. In: *Proceedings of the 2010 Workshop on Agents in Real Time and Environment*, pp. 35–44.
- Eagle, J.N., 1984. The approximate solution of a simple constrained search path moving target problem using moving horizon policies. *Monterey. Naval Postgraduate School, California*.
- Eagle, J.N., 1984. The optimal search for a moving target when the search path is constrained. *Oper. Res.* 32 (5), 1107–1115.
- Eagle, J.N., Yee, J.R., 1990. An optimal branch-and-bound procedure for the constrained path, moving target search problem. *Oper. Res.* 38 (1), 110–114.
- Foraker, J., Royset, J.O., Kaminer, I., 2016. Search-trajectory optimization: part I, formulation and theory. *J. Optim. Theory Appl.* 169 (2), 530–549.
- Foraker, J., Royset, J.O., Kaminer, I., 2016. Search-trajectory optimization: part II, algorithms and computations. *J. Optim. Theory Appl.* 169 (2), 550–567.
- Foraker, J.C., 2011. *Optimal Search for Moving Targets in Continuous Time and Space using Consistent Approximations*. Naval Postgraduate School, Monterey, CA Ph.D. thesis.
- Frank, M., Wolfe, P., 1956. An algorithm for quadratic programming. *Nav. Res. Logist.* 3 (1–2), 95–110.
- Furukawa, T., Durrant-Whyte, H., Lavis, B., 2007. The element-based method – theory and its application to Bayesian search and tracking. In: *Proceedings of the 2007 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 2807–2812.
- Hibey, J.L., 1982. Control-theoretic approach to optimal search for a class of Markovian targets. In: *Proceedings of the 1982 American Control Conference*, IEEE, pp. 705–709.
- Hohzaki, R., Iida, K., 1997. Optimal strategy of route and look for the path constrained search problem with reward criterion. *Eur. J. Oper. Res.* 100 (1), 236–249.
- Hong, S.P., Cho, S.J., Park, M.J., 2009. A pseudo-polynomial heuristic for path-constrained discrete-time Markovian-target search. *Eur. J. Oper. Res.* 193 (2), 351–364.
- Hong, S.P., Cho, S.J., Park, M.J., Lee, M.G., 2009. Optimal search-relocation trade-off in Markovian-target searching. *Comput. Oper. Res.* 36 (6), 2097–2104.
- Hu, J., Xie, L., Xu, J., 2012. Vision-based multi-agent cooperative target search. In: *Proceedings of the 2012 International Conference on Control Automation Robotics Vision*, pp. 895–900.
- Kratzke, T., Stone, L., Frost, J., 2010. Search and rescue optimal planning system. In: *Proceedings of the 13th Conference on Information Fusion*, pp. 1–8.
- Kullback, S., Leibler, R.A., 1951. On information and sufficiency. *Ann. Math. Stat.* 79–86.
- Lanillos, P., Besada-Portas, E., Pajares, G., Ruz, J., 2012. Minimum time search for lost targets using cross entropy optimization. In: *Proceedings of the 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 602–609.
- Lau, H., Huang, S., Dissanayake, G., 2008. Discounted MEAN bound for the optimal searcher path problem with non-uniform travel times. *Eur. J. Oper. Res.* 190 (2), 383–397.
- MacKay, D.J., 2002. *Information theory. Inference & Learning Algorithms*. Cambridge University Press, New York, NY, USA.
- Martins, G.H., 1993. *A New Branch and Bound Procedure for Computing Optimal Search Paths*. Naval Postgraduate School, Monterey, CA Master's thesis.
- Meng, W., He, Z., Su, R., Yadav, P.K., Teo, R., Xie, L., 2016. Decentralized multi-UAV flight autonomy for moving convoys search and track. *IEEE Trans. Control Syst. Technol.* (99) 1–8.
- Morin, M., Lamontagne, L., Abi-zeid, I., Lang, P., Maupin, P., 2009. The optimal searcher path problem with a visibility criterion in discrete time and space. In: *Proceedings of the 12th International Conference on Information Fusion*, pp. 2217–2224.
- Morin, M., Lamontagne, L., Abi-Zeid, I., Maupin, P., 2010. The ant search algorithm: An ant colony optimization algorithm for the optimal searcher path problem with visibility. *Can. Conf. Artif. Intell., Springer*, pp. 196–207.
- Morin, M., Papillon, A.P., Abi-Zeid, I., Laviolette, F., Quimper, C.G., 2012. Constraint programming for path planning with uncertainty. In: Milano, M. (Ed.), *Principles and Practice of Constraint Programming*, Lecture Notes in Computer Science. Springer, Berlin Heidelberg, pp. 988–1003.
- Norris, J., 1999. *Markov Chains*, No. 2008 in Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press.
- Nygårds, J., Skoglar, P., Karlholm, J., Ulvklo, M., Björström, R., 2005. Towards Concurrent Sensor and Path Planning – a Survey of Planning Methods Applicable to UAV Surveillance. *Sensor Technology Scientific Report ISSN*.
- Ohsumi, A., 1984. Stochastic control with searching a randomly moving target. In: *Proceedings of the 1984 American Control Conference*, pp. 500–504.
- Peng, H., Huo, M.L., Liu, Z.Z., Xu, W., 2015. Simulation analysis of cooperative target search strategies for multiple UAVs. In: *Control and Decision Conference (CCDC)*, 2015 27th Chinese. IEEE, pp. 4855–4859.
- Perez-Carabaza, S., Bermudez-Ortega, J., Besada-Portas, E., Lopez-Orozco, J.A., de la Cruz, J.M., 2017. A multi-UAV minimum time search planner based on ACOR. In: *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO '17*. ACM, New York, NY, USA, pp. 35–42.
- Perez-Carabaza, S., Besada-Portas, E., Lopez-Orozco, J.A., de la Cruz, J.M., 2016. A Real World Multi-UAV Evolutionary Planner for Minimum Time Target Detection. In: *Proceedings of the Genetic and Evolutionary Computation Conference 2016, GECCO '16*. ACM, New York, NY, USA, pp. 981–988.
- Raap, M., Meyer-Nieberg, S., Pickl, S., Szifkovits, M., 2017. Aerial vehicle search-path optimization: a novel method for emergency operations. *J. Optim. Theory Appl.* 172 (3), 965–983.
- Raap, M., Szifkovits, M., Pickl, S., 2017. Trajectory optimization under kinematical constraints for moving target search. *Comput. Oper. Res.* 88 (Supplement C), 324–331.
- Royset, J.O., Sato, H., 2010. Route optimization for multiple searchers. *Nav. Res. Logist.* 57 (8), 701–717.
- Sato, H., 2008. *Path Optimization for Single and Multiple Searchers: Models and Algorithms*. Naval Postgraduate School Ph.D. thesis.
- Sato, H., Royset, J.O., 2010. Path optimization for the resource-constrained searcher. *Nav. Res. Logist.* 57 (5), 422–440.
- Stewart, T., 1979. Search for a moving target when searcher motion is restricted. *Comput. Oper. Res.* 6 (3), 129–140.
- Stewart, T.J., 1985. Optimizing search with positive information feedback. *Nav. Res. Logist.* 32 (2), 263–274.
- Stone, L.D., Royset, J.O., Washburn, A.R., 2016. *Optimal Search for Moving Targets*. Springer.
- Stewart, T., 1980. Experience with a branch-and-bound algorithm for constrained searcher motion. In: *Search theory and applications*. Springer, pp. 247–253.
- Thomas, L.C., Eagle, J.N., 1995. Criteria and approximate methods for path-constrained moving-target search problems. *Nav. Res. Logist.* 42 (1), 27–38.
- Walton, C.L., Gong, Q., Kaminer, I., Royset, J.O., 2014. Optimal motion planning for searching for uncertain targets. In: *World Congress*, Vol. 19, pp. 8977–8982.

- Washburn, A.R., 1983. Search for a moving target: the FAB algorithm. *Oper. Res.* 31 (4), 739–751.
- Washburn, A.R., 1995. Branch and bound methods for search problems. Monterey. Naval Postgraduate School, California.
- Washburn, A.R., 1998. Branch and bound methods for a search problem. *Nav. Res. Logist.* 45 (3), 243–257.
- Wong, E.M., Bourgault, F., Furukawa, T., 2005. Multi-vehicle Bayesian search for multiple lost targets. In: *Proceedings of the 2005 IEEE International Conference on Robotics and Automation*, pp. 3169–3174.
- Xiao, X., Dong, Z., Wu, J., Duan, H., 2012. A cooperative approach to multiple uavs searching for moving targets based on a hybrid of virtual force and receding horizon. In: *IEEE 10th International Conference on Industrial Informatics*, IEEE, pp. 1228–1233.